

Team Note of BunCua Team

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1 Template

2 Data Structure

2.1 Segment Tree Walk

```

/* Find first element in range [u, v] have value >= val */
int findPos(int id, int l, int r, int u, int v, ll val) {
    if(l > v || r < u) return n + 1;
    if(st[id] < val) return n + 1;
    if(l == r) return l;
    push(id);
    int mid = (l + r) / 2;

```

```

    int res = findPos(id * 2, l, mid, u, v, val);
    if(res == n + 1) {
        res = findPos(id * 2 + 1, mid + 1, r, u, v, val);
    }
    return res;
}

```

2.2 Persistent Segment Tree

```

const int N = 3e5 + 5;
const int LOG = 20;
const int MAX_NODES = N * 40;

struct Node {
    int left, right, v;
    Node(int _left = 0, int _right = 0, int _v = 0) {
        left = _left;
        right = _right;
        v = _v;
    }
} st[MAX_NODES];

int cnt = 0;
int ver[N], nver = 0;

void refine(int id) {
    st[id].v = st[st[id].left].v + st[st[id].right].v;
}

int build(int l, int r) {
    if (l == r) {
        cnt++;
        st[cnt] = Node(0, 0, 0);
        return cnt;
    }
    int mid = (l + r) >> 1;
    int cur = ++cnt;
    st[cur].left = build(l, mid);
    st[cur].right = build(mid + 1, r);
    refine(cur);
    return cur;
}

int update(int id, int l, int r, int u, int x) {
    if (l == r) {
        cnt++;
        st[cnt] = Node(0, 0, st[id].v + x);
        return cnt;
    }
    int mid = (l + r) >> 1;
    int cur = ++cnt;
    if (u <= mid) {
        st[cur].left = update(st[id].left, l, mid, u, x);
        st[cur].right = st[id].right;
    } else {
        st[cur].left = st[id].left;
        st[cur].right = update(st[id].right, mid + 1, r, u, x);
    }
    refine(cur);
    return cur;
}

int get(int id, int l, int r, int u, int v) {
    if (l > v || r < u) return 0;
    if (l >= u && r <= v) return st[id].v;
    int mid = (l + r) >> 1;
    return get(st[id].left, l, mid, u, v) + get(st[id].right,
        mid + 1, r, u, v);
}

```

```

void add_version(int l, int r, int u, int x) {
    nver++;
    ver[nver] = update(ver[nver - 1], l, r, u, x);
}
/*
    USAGE EXAMPLE (Solving Distinct Values in Range - DQUERY):

    // 1. Initialize base tree
    cnt = nver = 0;
    ver[0] = build(1, n);

    // 2. Build versions tracking the latest occurrence of each
    value
    for (int i = 1; i <= n; i++) {
        // If value appeared before, remove its previous position
        if (mark[a[i]] > 0) {
            ver[nver] = update(ver[nver], 1, n, mark[a[i]], -1);
        }

        // Add current position
        nver++;
        ver[nver] = update(ver[nver - 1], 1, n, i, 1);

        iver[i] = ver[nver]; // iver[i] stores the version for
        prefix i
        mark[a[i]] = i;
    }

    // 3. Query distinct values in [L, R]
    int ans = get(iver[R], 1, n, L, R);
*/

```

2.3 Trie

```

const int LOG_MAX = 29;
const int MAX_NODES = 3e5 * 32;

struct Node {
    int c[2];
    int cnt;

    Node() {
        c[0] = c[1] = 0;
        cnt = 0;
    }
};

Node trie[MAX_NODES];
int node_cnt = 0;

void init_trie() {
    for (int i = 0; i <= node_cnt; i++) {
        trie[i] = Node();
    }
    node_cnt = 0;
}

void insert(int val) {
    int cur = 0;
    for (int i = LOG_MAX; i >= 0; i--) {
        int x = (val >> i) & 1;
        if (trie[cur].c[x] == 0) {
            trie[cur].c[x] = ++node_cnt;
        }
        cur = trie[cur].c[x];
        trie[cur].cnt++;
    }
}

bool search(int val) {
    int cur = 0;
    for (int i = LOG_MAX; i >= 0; i--) {
        int x = (val >> i) & 1;
        cur = trie[cur].c[x];
        if (cur == 0 || trie[cur].cnt == 0) {
            return false;
        }
    }
    return trie[cur].cnt > 0;
}

```

```

int count_occurrences(int val) {
    int cur = 0;
    for (int i = LOG_MAX; i >= 0; i--) {
        int x = (val >> i) & 1;
        cur = trie[cur].c[x];
        if (cur == 0 || trie[cur].cnt == 0) {
            return 0;
        }
    }
    return trie[cur].cnt;
}

void erase(int val) {
    if (!search(val)) return;
    int cur = 0;
    for (int i = LOG_MAX; i >= 0; i--) {
        int x = (val >> i) & 1;
        cur = trie[cur].c[x];
        trie[cur].cnt--;
    }
}

void update(int old_val, int new_val) {
    if (!search(old_val)) return;
    erase(old_val);
    insert(new_val);
}

```

2.4 Persistent Trie

```

const int LOG_MAX = 29;
const int N = 1e5 + 5;
const int MAX_NODES = N * 32;

int t[MAX_NODES][2];
int num[MAX_NODES];
int root[N];
int cnt = 0;

void init_trie() {
    for (int i = 0; i <= cnt; i++) {
        t[i][0] = t[i][1] = num[i] = 0;
    }
    cnt = 0;
    root[0] = 0;
}

int newnode() {
    cnt++;
    t[cnt][0] = t[cnt][1] = num[cnt] = 0;
    return cnt;
}

int add_version(int prev_root, int val) {
    int cur_root = newnode();
    int cur = cur_root;
    int old = prev_root;
    for (int i = LOG_MAX; i >= 0; i--) {
        int x = (val >> i) & 1;
        t[cur][0] = t[old][0];
        t[cur][1] = t[old][1];
        t[cur][x] = newnode();
        cur = t[cur][x];
        old = t[old][x];
        t[cur][0] = t[old][0];
        t[cur][1] = t[old][1];
        num[cur] = num[old] + 1;
    }
    return cur_root;
}

// 4. Query Maximum XOR with 'val' using combinations of
versions.
// Example: Query on a tree path from u to v with LCA p,
parent of p is pp.
// The effective count of a node is: num[x] + num[y] - num[p]
- num[pp].
int query_max_xor(int rootX, int rootY, int rootP, int rootPP,
int val) {
    int res = 0;
    int x = rootX, y = rootY, p = rootP, pp = rootPP;

```

```

for (int i = LOG_MAX; i >= 0; i--) {
    int v = (val >> i) & 1;
    int count_opposite = num[t[x][v ^ 1]] + num[t[y][v ^ 1]]
        - num[t[p][v ^ 1]] - num[t[pp][v ^ 1]];
    if (count_opposite > 0) {
        res |= (1 << i);
        x = t[x][v ^ 1];
        y = t[y][v ^ 1];
        p = t[p][v ^ 1];
        pp = t[pp][v ^ 1];
    } else {
        x = t[x][v];
        y = t[y][v];
        p = t[p][v];
        pp = t[pp][v];
    }
}
return res;
}

/*
USAGE EXAMPLE (Max XOR in an array from L to R):

// 1. Build prefix versions
init_trie();
for (int i = 1; i <= n; i++) {
    root[i] = add_version(root[i - 1], a[i]);
}

// 2. Query Max XOR with 'val' in subarray [L, R]
// For arrays, the effective tree is root[R] - root[L-1].
// You can use a modified query function that only takes 2
roots:
// count_opposite = num[t[rootR][v ^ 1]] -
num[t[rootL_minus_1][v ^ 1]];
*/

```

2.5 Fenwick Tree

```

struct Fenwick {
    int n;
    vector<long long> bit;

    Fenwick(int _n) {
        n = _n;
        bit.assign(n + 1, 0);
    }

    void update(int idx, long long delta) {
        for (; idx <= n; idx += idx & -idx) {
            bit[idx] += delta;
        }
    }

    long long query(int idx) {
        long long sum = 0;
        for (; idx > 0; idx -= idx & -idx) {
            sum += bit[idx];
        }
        return sum;
    }

    long long query(int l, int r) {
        if (l > r) return 0;
        return query(r) - query(l - 1);
    }

    // O(log N) Bit Jumping: Finds the FIRST index where prefix
    sum >= val.
    // Note: This only works if all elements in the BIT are
    non-negative!
    int lower_bound(long long val) {
        if (val <= 0) return 1;
        int pos = 0;
        long long sum = 0;
        for (int i = __lg(n); i >= 0; i--) {
            int next_pos = pos + (1 << i);
            if (next_pos <= n && sum + bit[next_pos] < val) {
                sum += bit[next_pos];
                pos = next_pos;
            }
        }
    }
}

```

```

}
// pos is the largest index such that prefix sum < val
// So pos + 1 is the first index such that prefix sum >=
val
return pos + 1;
}
};

```

2.6 Fenwick Tree 2D

```

/*
Idea:
- 1-based indexing for a grid of size N x M.
- When we add `v` to a 2D difference array at (i, j), it
contributes to
the prefix sum S(X, Y) (where X >= i and Y >= j) by:
v * (X - i + 1) * (Y - j + 1)
= v * [ (X+1)(Y+1) - (X+1)*j - (Y+1)*i + i*j ]
- To maintain this, we need 4 2D BITs to store the sum of:
1. v
2. v * i
3. v * j
4. v * i * j
*/

struct Fenwick2D {
    int n, m;
    vector<vector<long long>> bit1, bit2, bit3, bit4;

    Fenwick2D(int _n, int _m) {
        n = _n;
        m = _m;
        bit1.assign(n + 2, vector<long long>(m + 2, 0));
        bit2.assign(n + 2, vector<long long>(m + 2, 0));
        bit3.assign(n + 2, vector<long long>(m + 2, 0));
        bit4.assign(n + 2, vector<long long>(m + 2, 0));
    }

    void add(int x, int y, long long val) {
        for (int i = x; i <= n; i += i & -i) {
            for (int j = y; j <= m; j += j & -j) {
                bit1[i][j] += val;
                bit2[i][j] += val * x;
                bit3[i][j] += val * y;
                bit4[i][j] += val * x * y;
            }
        }
    }

    void update(int x1, int y1, int x2, int y2, long long val) {
        add(x1, y1, val);
        add(x2 + 1, y1, -val);
        add(x1, y2 + 1, -val);
        add(x2 + 1, y2 + 1, val);
    }

    // Internal Prefix Query: gets the sum of submatrix from (1,
    1) to (x, y)
    long long query_prefix(int x, int y) {
        long long res = 0;
        for (int i = x; i > 0; i -= i & -i) {
            for (int j = y; j > 0; j -= j & -j) {
                res += bit1[i][j] * (x + 1) * (y + 1)
                    - bit2[i][j] * (y + 1)
                    - bit3[i][j] * (x + 1)
                    + bit4[i][j];
            }
        }
        return res;
    }

    // Range Query: gets the sum of submatrix from (x1, y1) to
    (x2, y2)
    long long query(int x1, int y1, int x2, int y2) {
        return query_prefix(x2, y2)
            - query_prefix(x1 - 1, y2)
            - query_prefix(x2, y1 - 1)
            + query_prefix(x1 - 1, y1 - 1);
    }
};

```

2.7 XOR Basis

```

/*
XOR BASIS (Linear Basis over GF(2))

Idea:
- Represents a set of numbers using a minimal set of
linearly independent
  numbers (basis).
- Any XOR sum of a subset in the original set can be formed
by a XOR sum
  of a subset of the basis.
- The size of the basis never exceeds the number of bits
(LOG_D).

Complexity:
- Insert / Check / Get Max: O(LOG_D)
- Merge two bases: O(LOG_D^2)
- Build RREF: O(LOG_D^2)
- K-th Smallest/Largest: O(LOG_D)
*/

#include <bits/stdc++.h>
using namespace std;

const int LOG_D = 60; // 60 for values up to 10^18, 30 for
values up to 10^9

struct XorBasis {
    long long basis[LOG_D];
    int sz; // Number of elements currently in the basis
    vector<long long> reduced_basis; // Stores the canonical
form for K-th queries

    XorBasis() {
        memset(basis, 0, sizeof(basis));
        sz = 0;
    }

    // 1. Insert a value. Returns true if successfully inserted
    // (i.e., the value is linearly independent from the current
basis).
    bool insert(long long x) {
        for (int i = LOG_D - 1; i >= 0; i--) {
            if ((x >> i) & 1) {
                if (!basis[i]) {
                    basis[i] = x;
                    sz++;
                    return true;
                }
                x ^= basis[i];
            }
        }
        return false; // x became 0, meaning it was already
spanned by the basis
    }

    // 2. Check if a value can be formed by XORing a subset of
the basis
    bool check(long long x) {
        for (int i = LOG_D - 1; i >= 0; i--) {
            if ((x >> i) & 1) {
                if (!basis[i]) return false;
                x ^= basis[i];
            }
        }
        return true;
    }

    // 3. Get the maximum possible XOR sum using the basis.
    long long get_max(long long start_val = 0) {
        long long res = start_val;
        for (int i = LOG_D - 1; i >= 0; i--) {
            if ((res ^ basis[i]) > res) {
                res ^= basis[i];
            }
        }
        return res;
    }

    // 4. Merge this basis with another basis
    void merge(const XorBasis& other) {

```

```

        for (int i = 0; i < LOG_D; i++) {
            if (other.basis[i]) {
                insert(other.basis[i]);
            }
        }
    }

    // K-TH LARGEST / SMALLEST QUERIES
    // Note: These functions only consider DISTINCT values
generated by the basis.
    // There are exactly 2^sz distinct values (including 0 from
the empty subset).

    // 5. Build Reduced Row Echelon Form (Canonical Form)
    // MUST be called after all inserts and BEFORE querying k-th
values!
    void build() {
        reduced_basis.clear();
        for (int i = LOG_D - 1; i >= 0; i--) {
            for (int j = i - 1; j >= 0; j--) {
                // If the j-th bit of basis[i] is set, use basis[j] to
turn it off
                if ((basis[i] >> j) & 1) {
                    basis[i] ^= basis[j];
                }
            }
        }
        // Collect non-zero elements into the reduced_basis array
(in ascending order)
        for (int i = 0; i < LOG_D; i++) {
            if (basis[i]) {
                reduced_basis.push_back(basis[i]);
            }
        }
    }

    // 6. Get the k-th smallest distinct XOR sum (1-indexed).
    // k = 1 returns 0 (the empty subset).
    // Returns -1 if k is out of bounds (k > 2^sz).
    long long get_kth_smallest(long long k) {
        long long total_combinations = 1LL << sz;
        if (k < 1 || k > total_combinations) return -1;

        long long rank = k - 1; // 0-indexed rank
        long long res = 0;
        for (int i = 0; i < sz; i++) {
            if ((rank >> i) & 1) {
                res ^= reduced_basis[i];
            }
        }
        return res;
    }

    // 7. Get the k-th largest distinct XOR sum (1-indexed).
    // k = 1 returns the absolute maximum possible XOR sum.
    // Returns -1 if k is out of bounds (k > 2^sz).
    long long get_kth_largest(long long k) {
        long long total_combinations = 1LL << sz;
        if (k < 1 || k > total_combinations) return -1;

        // The k-th largest is equivalent to the
(total_combinations - k)-th smallest
        long long rank = total_combinations - k;
        long long res = 0;
        for (int i = 0; i < sz; i++) {
            if ((rank >> i) & 1) {
                res ^= reduced_basis[i];
            }
        }
        return res;
    }
};

/*
USAGE EXAMPLE:

signed main() {
    int n = 3;
    long long a[] = {2, 4, 6}; // 2 (010), 4 (100), 6 (110)

    XorBasis xb;

```

```

    for (int i = 0; i < n; i++) xb.insert(a[i]);

    // 1. Build the canonical form before calling k-th
    queries
    xb.build();

    // The distinct XOR sums generated are: 0, 2, 4, 6
    // Since 6 is just  $2^2$ , the basis size (sz) will be 2.
    (2^2 = 4 combinations)

    cout << "Max XOR sum: " << xb.get_max() << "\n"; // 6

    cout << "1st smallest: " << xb.get_kth_smallest(1) <<
    "\n"; // 0
    cout << "2nd smallest: " << xb.get_kth_smallest(2) <<
    "\n"; // 2

    cout << "1st largest: " << xb.get_kth_largest(1) << "\n";
    // 6
    cout << "3rd largest: " << xb.get_kth_largest(3) << "\n";
    // 2

    return 0;
}
*/

```

2.8 DSU Rollback

```

/*
DISJOINT SET UNION WITH ROLLBACK
Idea:
- Supports union of two sets and rolling back to a previous
state.
- STRICT RULE: NO PATH COMPRESSION! Path compression
destroys the tree
  structure, making it impossible to undo operations
  efficiently.
- We only use "Union by Size". This guarantees that the
tree depth
  never exceeds  $O(\log N)$ .
Complexity:
- find_set:  $O(\log N)$ 
- union_sets:  $O(\log N)$ 
- rollback:  $O(1)$  per undone operation
*/
struct DSU_Rollback {
    vector<int> parent, sz;
    int components;

    // Stores the history of unions as pairs {child, parent}
    vector<pair<int, int>> history;

    DSU_Rollback(int n) {
        parent.assign(n + 1, 0);
        sz.assign(n + 1, 1);
        components = n;
        for (int i = 1; i <= n; i++) {
            parent[i] = i;
        }
    }

    // Find set WITHOUT path compression
    int find_set(int v) {
        while (v != parent[v]) {
            v = parent[v];
        }
        return v;
    }

    // Union two sets by size and record the operation
    bool union_sets(int a, int b) {
        a = find_set(a);
        b = find_set(b);
        if (a == b) return false;

        // Union by size: always attach the smaller tree (a) to
        the larger one (b)
        if (sz[a] > sz[b]) {
            swap(a, b);
        }
    }
}

```

```

// Perform union
parent[a] = b;
sz[b] += sz[a];
components--;

// Record the operation: 'a' became a child of 'b'
history.push_back({a, b});
return true;
}

// Take a snapshot of the current state (returns the size of
the history stack)
int snapshot() {
    return history.size();
}

// Rollback operations until the history stack size is equal
to `checkpoint`
void rollback(int checkpoint) {
    while (history.size() > checkpoint) {
        int a = history.back().first;
        int b = history.back().second;
        history.pop_back();

        // Undo the union operation
        sz[b] -= sz[a];
        parent[a] = a;
        components++;
    }
}
};

/*
USAGE EXAMPLE (Dynamic Connectivity / Divide & Conquer on
Time):

signed main() {
    int n = 5;
    DSU_Rollback dsu(n);

    dsu.union_sets(1, 2);
    dsu.union_sets(3, 4);
    cout << "Components: " << dsu.components << "\n"; //
Output: 3

    // 1. Save the current state
    int checkpoint = dsu.snapshot();

    // 2. Do some temporary unions
    dsu.union_sets(2, 3);
    dsu.union_sets(4, 5);
    cout << "Components after temp unions: " <<
    dsu.components << "\n"; // Output: 1

    // 3. Rollback to the saved state
    dsu.rollback(checkpoint);

    // The state is perfectly restored!
    cout << "Components after rollback: " << dsu.components
    << "\n"; // Output: 3

    return 0;
}
*/

```

2.9 Ordered Set

```

/*
GNU PBDS ORDERED SET (Policy-Based Data Structure)

Idea:
- A balanced binary search tree (Red-Black Tree) with order
statistics.
- Works exactly like std::set but with two additional  $O(\log
N)$  functions:
    1. find_by_order(k): returns an iterator to the k-th
    smallest element (0-indexed).
    2. order_of_key(x): returns the number of elements
    strictly smaller than x.

IMPORTANT FOR MULTISSET:

```

```

- Do NOT use `less_equal<T>` for multiset if you need to
use `.erase()`.
  It will delete ALL occurrences of the value, or break the
  tree structure.
- The best and safest way to implement an ordered_multiset
is to use
  `pair<T, int>` where the second element is a unique
  counter/ID.
*/
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

using namespace std;
using namespace __gnu_pbds;

// 1. Ordered Set (Distinct elements only)
template <typename T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,
tree_order_statistics_node_update>;

// 2. Ordered Multiset (Allows duplicates safely by using
pair<T, int>)
template <typename T>
using ordered_multiset = tree<pair<T, int>, null_type,
less<pair<T, int>>, rb_tree_tag,
tree_order_statistics_node_update>;

/*
  USAGE EXAMPLE:

  signed main() {
    //
    -----
    // EXAMPLE 1: ORDERED SET (Distinct Elements)
    //
    -----
    ordered_set<int> os;
    os.insert(10);
    os.insert(30);
    os.insert(20);
    os.insert(30); // Ignored, as 30 is already present

    // find_by_order(k): Returns iterator to the k-th
    smallest element (0-indexed)
    cout << *os.find_by_order(0) << "\n"; // Output: 10
    cout << *os.find_by_order(2) << "\n"; // Output: 30

    // order_of_key(x): Returns the number of elements
    STRICTLY smaller than x
    cout << os.order_of_key(25) << "\n"; // Output: 2 (10
    and 20)
    cout << os.order_of_key(20) << "\n"; // Output: 1 (only
    10)

    os.erase(20); // Erases 20 normally

    //
    -----
    // EXAMPLE 2: ORDERED MULTISSET (With Duplicates)
    //
    -----
    ordered_multiset<int> oms;
    int timer = 0; // Unique ID counter to distinguish
    duplicates

    // Insert: use pair {value, ++timer}
    oms.insert({10, ++timer});
    oms.insert({20, ++timer});
    oms.insert({20, ++timer});
    oms.insert({30, ++timer});

    // order_of_key: just pass {value, -1} to get count of
    elements < value
    cout << oms.order_of_key({20, -1}) << "\n"; // Output: 1
    (only {10})
    cout << oms.order_of_key({30, -1}) << "\n"; // Output: 3
    ({10}, {20}, {20})

    // find_by_order works exactly the same, but returns a
    pair

```

```

    cout << oms.find_by_order(1)->first << "\n"; // Output:
    20
    cout << oms.find_by_order(2)->first << "\n"; // Output:
    20

    // Erase one occurrence of 20:
    // Find its iterator using lower_bound with -1, then
    erase the iterator
    auto it = oms.lower_bound({20, -1});
    if (it != oms.end() && it->first == 20) {
        oms.erase(it);
    }

    return 0;
  }
*/

```

2.10 Line Container

```

/*
  DYNAMIC CONVEX HULL TRICK (LINE CONTAINER)
  Idea:
  - Maintains the upper envelope of a set of lines  $y = kx + m$ .
  - Allows dynamically adding lines and querying the MAXIMUM
  y value at
    a given x in  $O(\log N)$  time.
  - For MINIMUM queries: insert(-k, -m) and negate the query
  result: -query(x).
  How it works (KACTL approach):
  - Uses std::multiset to keep lines sorted by slope 'k'.
  - Maintains the intersection point 'p' with the next line.
  - Uses C++14 heterogeneous lookup (less<>) to binary search
  by 'x' value
    using the 'p' attribute.
  Complexity:
  - Add line: Amortized  $O(\log N)$ 
  - Query:  $O(\log N)$ 
*/
const long long LINF = 2e18;
struct Line {
    mutable long long k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(long long x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    long long div(long long a, long long b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }

    bool isect(iterator x, iterator y) {
        if (y == end()) { x->p = LINF; return false; }
        if (x->k == y->k) {
            x->p = x->m > y->m ? LINF : -LINF;
        } else {
            x->p = div(y->m - x->m, x->k - y->k);
        }
        return x->p >= y->p;
    }

    // Adds a line  $y = kx + m$  to the envelope
    void add(long long k, long long m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p) {
            isect(x, erase(y));
        }
    }

    // Queries the maximum y value at a given x
    long long query(long long x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};

/*
  USAGE EXAMPLE:

```

```
signed main() {
    LineContainer lc;
    //
    -----
    // EXAMPLE 1: MAXIMUM QUERIES (Default behavior)
    //
    -----
    // Add lines: y = 2x + 3, y = 1x + 10, y = 3x - 5
    lc.add(2, 3);
    lc.add(1, 10);
    lc.add(3, -5);

    cout << "Max at x=2: " << lc.query(2) << "\n"; // y =
    1*(2) + 10 = 12
    cout << "Max at x=10: " << lc.query(10) << "\n"; // y =
    3*(10) - 5 = 25

    //
    -----
    // EXAMPLE 2: MINIMUM QUERIES (Trick)
    //
    -----
    LineContainer min_lc;

    // We want to find MINIMUM of: y = 2x + 3, y = 1x + 10, y
    = 3x - 5
    // Trick: Insert (-k, -m)
    min_lc.add(-2, -3);
    min_lc.add(-1, -10);
    min_lc.add(-3, 5); // -(-5) = 5

    // Trick: Negate the query result
    cout << "Min at x=2: " << -min_lc.query(2) << "\n"; // y =
    2*(2) + 3 = 7
    cout << "Min at x=10: " << -min_lc.query(10) << "\n"; //
    y = 1*(10) + 10 = 20

    return 0;
}
*/
```

2.11 Deque CHT

```
struct Line {
    long long k, m;
    long long get(long long x) { return k * x + m; }
};

struct DequeCHT {
    deque<Line> dq;

    long long floor_div(long long a, long long b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }

    long long intersect(Line a, Line b) {
        return floor_div(a.m - b.m, b.k - a.k);
    }

    // Adds a line y = k*x + m.
    // IMPORTANT: Slopes (k) must be inserted in non-decreasing
    order!
    void add(long long k, long long m) {
        Line L = {k, m};

        if (!dq.empty() && dq.back().k == k) {
            if (m <= dq.back().m) return;
            dq.pop_back();
        }

        while (dq.size() >= 2) {
            Line l1 = dq[dq.size() - 2];
            Line l2 = dq.back();
            if (intersect(l1, l2) >= intersect(l2, L)) {
                dq.pop_back();
            } else {
                break;
            }
        }
        dq.push_back(L);
    }
};
```

```
// 1. O(1) Query: ONLY use when query values 'x' are
monotonically INCREASING!
long long query_increasing_x(long long x) {
    assert(!dq.empty());
    while (dq.size() >= 2 && dq[0].get(x) <= dq[1].get(x)) {
        dq.pop_front();
    }
    return dq[0].get(x);
}

// 2. O(log N) Query: Use when 'x' values are RANDOM / NOT
SORTED
long long query_binary_search(long long x) {
    assert(!dq.empty());
    int l = 0, r = dq.size() - 2;
    int ans = dq.size() - 1;

    while (l <= r) {
        int mid = l + (r - l) / 2;
        if (intersect(dq[mid], dq[mid + 1]) < x) {
            l = mid + 1;
        } else {
            ans = mid;
            r = mid - 1;
        }
    }
    return dq[ans].get(x);
}
};
```

3 Graphs

3.1 2 SAT

```
/*
- For a boolean variable x, we create two nodes: x (1..n)
and NOT(x) (n+1..2n).
- A clause (x OR y) is equivalent to: (!x => y) and (!y =>
x).
- If x and NOT(x) are in the same Strongly Connected
Component (SCC),
it's a contradiction -> NO solution.
- We assign TRUE to x if its SCC is topologically sorted
AFTER NOT(x)'s SCC.
(Tarjan assigns SCC IDs in reverse topological order, so
col[x] < col[NOT(x)])
*/

const int N = 2e4 + 5;

int n;
vector<int> G[2 * N];
int num[2 * N], low[2 * N], cnt;
int col[2 * N], ct;
vector<int> stk;
int ans[N];

int NOT(int x) {
    return (x <= n ? x + n : x - n);
}

void addedge(int x, int y) {
    G[NOT(x)].push_back(y);
    G[NOT(y)].push_back(x);
}

void tarjan(int u) {
    num[u] = low[u] = ++cnt;
    stk.push_back(u);
    for (int v : G[u]) {
        if (!col[v]) {
            if (num[v]) {
                low[u] = min(low[u], num[v]);
            } else {
                tarjan(v);
                low[u] = min(low[u], low[v]);
            }
        }
    }
}
```

```

    if (num[u] == low[u]) {
        ct++;
        while (true) {
            int v = stk.back();
            stk.pop_back();
            col[v] = ct;
            if (v == u) break;
        }
    }
}

bool solve() {
    rep(i, 1, 2 * n) {
        if (!num[i]) tarjan(i);
    }
    rep(i, 1, n) {
        if (col[i] == col[NOT(i)]) return false;
        ans[i] = (col[i] < col[NOT(i)]);
    }
    return true;
}

```

3.2 LCA O(1)

```

const int N = 1e5 + 5;
const int LOG = 18;

vector<int> G[N];
int tin[N], cnt, rn[N];
int euler[2 * N], ct, first[N];
int depth[N];
int rmq[LOG][2 * N];

void dfs(int u, int p = 0) {
    tin[u] = ++cnt;
    rn[cnt] = u;
    euler[++ct] = tin[u];
    first[tin[u]] = ct;
    for (int v : G[u]) {
        if (v == p) continue;
        depth[v] = depth[u] + 1;
        dfs(v, u);
        euler[++ct] = tin[u];
    }
}

void build_lca() {
    for (int i = 1; i <= ct; i++) {
        rmq[0][i] = euler[i];
    }
    for (int j = 1; j < LOG; j++) {
        for (int i = 1; i <= ct - (1 << (j - 1)); i++) {
            rmq[j][i] = min(rmq[j - 1][i], rmq[j - 1][i + (1 << (j - 1))]);
        }
    }
}

int lca(int x, int y) {
    x = first[tin[x]];
    y = first[tin[y]];
    if (x > y) swap(x, y);
    int k = 31 - __builtin_clz(y - x + 1);
    int min_tin = min(rmq[k][x], rmq[k][y - (1 << k) + 1]);
    return rn[min_tin];
}

```

3.3 Centroid Decomposition

```

const int N = 3e5 + 5;
vector<int> G[N];
int sz[N];
bool deleted[N];
int centroid_par[N];

void get_sz(int u, int par) {
    sz[u] = 1;
    for (int v : G[u]) {
        if (v != par && !deleted[v]) {
            get_sz(v, u);

```

```

            sz[u] += sz[v];
        }
    }
}

int get_centroid(int u, int par, int total_nodes) {
    for (int v : G[u]) {
        if (v != par && !deleted[v]) {
            if (sz[v] > total_nodes / 2) {
                return get_centroid(v, u, total_nodes);
            }
        }
    }
    return u;
}

void solve_centroid(int centroid) {
    // TODO: Add your logic here (e.g., DFS to collect all
    // distances from the
    // centroid to the nodes in its component, then count valid
    // pairs).
    //
    // Example structure:
    // for (int v : G[centroid]) {
    //     if (!deleted[v]) {
    //         // 1. DFS(v) to query answer with currently stored
    //         // paths
    //         // 2. DFS(v) again to add new paths to our data
    //         // structure
    //     }
    // }
    // // 3. Clear the data structure for the next centroid
}

void build_centroid(int u, int p = -1) {
    get_sz(u, -1);
    int centroid = get_centroid(u, -1, sz[u]);
    deleted[centroid] = true;
    centroid_par[centroid] = p;
    solve_centroid(centroid);
    for (int v : G[centroid]) {
        if (!deleted[v]) {
            build_centroid(v, centroid);
        }
    }
}

```

3.4 Heavy Light Decomposition

```

const int N = 2e5 + 5;
vector<int> G[N];
int sz[N], son[N], depth[N], pre[N];
int in[N], cnt, top[N];

void dfs1(int u, int par) {
    sz[u] = 1;
    son[u] = 0;
    pre[u] = par;
    for (int v : G[u]) {
        if (v != par) {
            depth[v] = depth[u] + 1;
            dfs1(v, u);
            sz[u] += sz[v];
            if (sz[v] > sz[son[u]]) {
                son[u] = v;
            }
        }
    }
}

void rebuild(int u, int topp) {
    in[u] = ++cnt;
    top[u] = topp;
    if (son[u]) {
        rebuild(son[u], topp);
    }
    for (int v : G[u]) {
        if (v != pre[u] && v != son[u]) {
            rebuild(v, v);
        }
    }
}

```

```

}

int query_path(int x, int y) {
    int res = 0;
    while (top[x] != top[y]) {
        if (depth[top[x]] < depth[top[y]]) {
            swap(x, y);
        }
        res = max(res, get(1, 1, n, in[top[x]], in[x]));
        x = pre[top[x]];
    }
    if (depth[x] > depth[y]) {
        swap(x, y);
    }
    res = max(res, get(1, 1, n, in[x], in[y]));
    return res;
}

void update_path(int x, int y, int val) {
    while (top[x] != top[y]) {
        if (depth[top[x]] < depth[top[y]]) {
            swap(x, y);
        }
        x = pre[top[x]];
    }
    if (depth[x] > depth[y]) {
        swap(x, y);
    }
}
*/
USAGE EXAMPLE:
dfs1(1, 1);
cnt = 0;
rebuild(1, 1);
// for(int i = 1; i <= n; i++) update(1, 1, n, in[i],
a[i]);
*/

```

3.5 Bellman Ford

```

const long long LINF = 1e18;

struct Edge {
    int u, v;
    long long w;
};

int n, m;
vector<Edge> edges;
long long dist[2005];
int parent[2005];

bool bellman_ford(int src) {
    for (int i = 1; i <= n; i++) {
        dist[i] = LINF;
        parent[i] = -1;
    }
    dist[src] = 0;
    for (int i = 1; i <= n - 1; i++) {
        bool updated = false;
        for (auto e : edges) {
            if (dist[e.u] < LINF && dist[e.u] + e.w < dist[e.v]) {
                dist[e.v] = dist[e.u] + e.w;
                parent[e.v] = e.u;
                updated = true;
            }
        }
        if (!updated) break;
    }
    bool has_negative_cycle = false;
    for (int i = 1; i <= n; i++) {
        for (auto e : edges) {
            if (dist[e.u] < LINF && dist[e.u] + e.w < dist[e.v]) {
                dist[e.v] = -LINF;
                has_negative_cycle = true;
            }
        }
    }
    return has_negative_cycle;
}

```

```

/*
HOW TO EXTRACT A NEGATIVE CYCLE (If needed)
If you only need to print ONE negative cycle (like CSES
Cycle Finding):
1. Keep an array `parent`.
2. Run Bellman-Ford for exactly N iterations (no need for
Phase 2 above).
3. On the N-th iteration, if an edge (u, v) is relaxed, `v`
is either in
    the negative cycle or reachable from it.
4. To find a node strictly inside the cycle, go back `N`
times using `parent`:
    int x = v;
    for (int i = 1; i <= n; i++) x = parent[x];
5. Now `x` is guaranteed to be in the cycle. Trace back
using `parent[x]`
    until you reach `x` again, push nodes to a vector, and
reverse it.
*/

```

3.6 Floyd

```

const long long LINF = 1e18;
const int N = 505;

long long dist[N][N];
int nxt[N][N];

void init(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            dist[i][j] = LINF;
            nxt[i][j] = -1;
        }
        dist[i][i] = 0;
    }
}

void add_edge(int u, int v, long long w) {
    if (w < dist[u][v]) {
        dist[u][v] = w;
        nxt[u][v] = v;
    }
}

void floyd_warshall(int n) {
    for (int k = 1; k <= n; k++) {
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                if (dist[i][k] < LINF && dist[k][j] < LINF) {
                    if (dist[i][k] + dist[k][j] < dist[i][j]) {
                        dist[i][j] = dist[i][k] + dist[k][j];
                        nxt[i][j] = nxt[i][k];
                    }
                }
            }
        }
    }

    // 4. (Optional) Propagate Negative Cycles
    // If node k is part of a negative cycle, any path going
    // through k
    // will also have a length of -Infinity.
    for (int k = 1; k <= n; k++) {
        if (dist[k][k] < 0) {
            for (int i = 1; i <= n; i++) {
                for (int j = 1; j <= n; j++) {
                    if (dist[i][k] < LINF && dist[k][j] < LINF) {
                        dist[i][j] = -LINF;
                    }
                }
            }
        }
    }

    // 5. Path Reconstruction
    // Returns the path from u to v, or an empty vector if no path
    // exists
    vector<int> get_path(int u, int v) {
        if (dist[u][v] == LINF) return {};
    }
}

```

```
vector<int> path;
for (int at = u; at != v; at = nxt[at][v]) {
    if (at == -1) return {};
    path.push_back(at);
}
path.push_back(v);
return path;
}
```

3.7 Ford Fulkerson

```
/*
    Idea:
    - Repeatedly find an augmenting path from Source (S) to Sink (T) using BFS.
    - Bottleneck of the path is the minimum remaining capacity among its edges.
    - Push the bottleneck flow through the path, and ADD it to the reverse edges (to allow "undoing" flow later).
    - Terminates when no more augmenting paths can be found.
    Complexity:  $O(V * E^2)$ . Safe for  $V \leq 1000$ ,  $E \leq 5000$ .
*/
const long long LINF = 1e18;

struct Edge {
    int u, v;
    long long cap, flow;
};

struct MaxFlow {
    int n;
    vector<Edge> edges;
    vector<vector<int>> adj;

    MaxFlow(int _n) {
        n = _n;
        adj.assign(n + 1, vector<int>());
    }

    // Add a directed edge from u to v with capacity `cap`.
    // If it's an undirected edge, set `is_directed = false`.
    void add_edge(int u, int v, long long cap, bool is_directed = true) {
        adj[u].push_back(edges.size());
        edges.push_back({u, v, cap, 0}); // Forward edge

        adj[v].push_back(edges.size());
        edges.push_back({v, u, is_directed ? 0 : cap, 0}); // Backward (residual) edge
    }

    // Find max flow from s to t
    long long max_flow(int s, int t) {
        long long total_flow = 0;
        vector<int> parent_edge(n + 1);

        while (true) {
            // 1. BFS to find an augmenting path
            fill(parent_edge.begin(), parent_edge.end(), -1);
            queue<int> q;
            q.push(s);
            parent_edge[s] = -2; // Mark source as visited

            while (!q.empty() && parent_edge[t] == -1) {
                int u = q.front();
                q.pop();

                for (int id : adj[u]) {
                    int v = edges[id].v;
                    // If v is not visited AND the edge has remaining capacity
                    if (parent_edge[v] == -1 && edges[id].cap - edges[id].flow > 0) {
                        parent_edge[v] = id; // Store the edge ID used to reach v
                        q.push(v);
                    }
                }
            }
        }
    }
}
```

```
// 2. If Sink (t) is not reached, no more paths exist -> break
if (parent_edge[t] == -1) break;

// 3. Find the bottleneck capacity along the path found by BFS
long long push = LINF;
int cur = t;
while (cur != s) {
    int id = parent_edge[cur];
    push = min(push, edges[id].cap - edges[id].flow);
    cur = edges[id].u;
}

// 4. Augment the flow along the path and update residual edges
cur = t;
while (cur != s) {
    int id = parent_edge[cur];
    edges[id].flow += push; // Add flow to forward edge
    edges[id ^ 1].flow -= push; // Subtract flow from backward edge (undo capability)
    cur = edges[id].u;
}

total_flow += push;
}
return total_flow;
}

// Get Min Cut: Returns a boolean array where visited[i] = true means
// vertex `i` belongs to the Source's component (Set S).
vector<bool> get_min_cut(int s) {
    vector<bool> visited(n + 1, false);
    queue<int> q;
    q.push(s);
    visited[s] = true;

    // BFS one last time on edges that are NOT fully saturated
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (int id : adj[u]) {
            int v = edges[id].v;
            if (!visited[v] && edges[id].cap - edges[id].flow > 0) {
                visited[v] = true;
                q.push(v);
            }
        }
    }
    return visited;
}

// Tracing: Print all edges that actually carry water in the final state
void print_flow_paths() {
    cout << "Edges with positive flow:\n";
    // We step by 2 to only check the original forward edges (0, 2, 4...)
    for (int i = 0; i < edges.size(); i += 2) {
        if (edges[i].flow > 0) {
            cout << edges[i].u << " -> " << edges[i].v << " : " << edges[i].flow << " / " << edges[i].cap << "\n";
        }
    }
}

/*
USAGE EXAMPLE:
signed main() {
    int n = 4; // 4 vertices
    MaxFlow mf(n);

    // add_edge(u, v, capacity)
    mf.add_edge(1, 2, 3);
}
```

```

mf.add_edge(1, 3, 2);
mf.add_edge(2, 3, 1);
mf.add_edge(2, 4, 2);
mf.add_edge(3, 4, 4);

int source = 1, sink = 4;
cout << "Max Flow: " << mf.max_flow(source, sink) <<
"\n";

// Print flow network details
mf.print_flow_paths();

// Find the Min Cut edges
vector<bool> in_source_set = mf.get_min_cut(source);
cout << "Min Cut Edges:\n";
for (int i = 0; i < mf.edges.size(); i += 2) {
    int u = mf.edges[i].u;
    int v = mf.edges[i].v;
    // An edge is in the Min Cut if it goes from the Source
    // Set to the Sink Set
    if (in_source_set[u] && !in_source_set[v]) {
        cout << u << " -> " << v << " (Capacity cut: " <<
        mf.edges[i].cap << ")\n";
    }
}

return 0;
}
*/

```

3.8 Dinic

```

/*
DINIC'S ALGORITHM (Max Flow & Min Cut)
Idea:
- Phase 1 (BFS): Build a "Level Graph". Assign levels to
nodes based on
    shortest path from Source (S). Only edges that go to the
    next level
    (level[v] == level[u] + 1) are considered valid.
- Phase 2 (DFS): Push as much flow as possible along the
Level Graph.
- DEAD-END OPTIMIZATION (ptr array): During DFS, if an edge
leads to a
    dead end (can't push any more flow), we advance the `ptr`
    so we NEVER
    visit this useless edge again in the current BFS phase.
Complexity:
- General Graph:  $O(V^2 * E)$ 
- Unit Network (e.g., Bipartite Matching):  $O(E * \sqrt{V})$ 
- Extremely fast in practice, much faster than
    Edmonds-Karp.
*/
const long long LINf = 1e18;

struct Edge {
    int u, v;
    long long cap, flow;
};

struct Dinic {
    int n;
    vector<Edge> edges;
    vector<vector<int>> adj;
    vector<int> level; // Stores the BFS level of each node
    vector<int> ptr; // Optimization: keeps track of the next
    edge to explore in DFS

    Dinic(int _n) {
        n = _n;
        adj.assign(n + 1, vector<int>());
        level.assign(n + 1, -1);
        ptr.assign(n + 1, 0);
    }

    void add_edge(int u, int v, long long cap, bool is_directed
= true) {
        adj[u].push_back(edges.size());
        edges.push_back({u, v, cap, 0});

        adj[v].push_back(edges.size());

```

```

        edges.push_back({v, u, is_directed ? 0 : cap, 0});
    }

    // BFS to build the Level Graph
    bool bfs(int s, int t) {
        fill(level.begin(), level.end(), -1);
        level[s] = 0;
        queue<int> q;
        q.push(s);

        while (!q.empty()) {
            int u = q.front();
            q.pop();
            for (int id : adj[u]) {
                int v = edges[id].v;
                // If not visited AND has remaining capacity
                if (level[v] == -1 && edges[id].cap - edges[id].flow >
0) {
                    level[v] = level[u] + 1;
                    q.push(v);
                }
            }
        }
        // Return true if the sink is reachable
        return level[t] != -1;
    }

    // DFS to push flow along the Level Graph
    long long dfs(int u, int t, long long pushed) {
        if (pushed == 0 || u == t) return pushed;

        // Note the reference `&cid = ptr[u]`. This is the core
        // optimization!
        // It prevents checking dead ends multiple times in the
        // same BFS phase.
        for (int& cid = ptr[u]; cid < adj[u].size(); ++cid) {
            int id = adj[u][cid];
            int v = edges[id].v;
            long long tr = edges[id].cap - edges[id].flow; //
            Remaining capacity

            // Only push flow to the exact next level
            if (level[u] + 1 != level[v] || tr == 0) continue;

            long long push = dfs(v, t, min(pushed, tr));
            if (push == 0) continue;

            edges[id].flow += push;
            edges[id ^ 1].flow -= push;
            return push;
        }
        return 0;
    }

    // Main function to calculate Max Flow
    long long max_flow(int s, int t) {
        long long flow = 0;
        while (bfs(s, t)) { // While a path exists
            fill(ptr.begin(), ptr.end(), 0); // Reset the dead-end
            pointers

            // Push as much flow as possible in the current Level
            // Graph
            while (long long pushed = dfs(s, t, LINf)) {
                flow += pushed;
            }
        }
        return flow;
    }

    // Min Cut Extraction: Nodes reachable from Source in the
    // residual graph
    vector<bool> get_min_cut(int s) {
        vector<bool> visited(n + 1, false);
        queue<int> q;
        q.push(s);
        visited[s] = true;

        while (!q.empty()) {
            int u = q.front();
            q.pop();

```

```

    for (int id : adj[u]) {
        int v = edges[id].v;
        if (!visited[v] && edges[id].cap - edges[id].flow > 0)
        {
            visited[v] = true;
            q.push(v);
        }
    }
    return visited; // visited[i] = true means 'i' is in the
Source set
}

// Tracing: Print edges that carry flow
void print_flow_paths() {
    cout << "Edges with positive flow:\n";
    for (int i = 0; i < edges.size(); i += 2) { // Check
forward edges only
        if (edges[i].flow > 0) {
            cout << edges[i].u << " -> " << edges[i].v
                << " : " << edges[i].flow << " / " <<
edges[i].cap << "\n";
        }
    }
}

};

/*
USAGE EXAMPLE:
signed main() {
    int n = 4; // 4 vertices
    Dinic dinic(n);

    // add_edge(u, v, capacity)
    dinic.add_edge(1, 2, 3);
    dinic.add_edge(1, 3, 2);
    dinic.add_edge(2, 3, 1);
    dinic.add_edge(2, 4, 2);
    dinic.add_edge(3, 4, 4);

    int source = 1, sink = 4;
    cout << "Max Flow: " << dinic.max_flow(source, sink) <<
"\n"; // Output: 5

    // Extract Min Cut
    vector<bool> in_source_set = dinic.get_min_cut(source);
    cout << "Min Cut Edges:\n";
    for (int i = 0; i < dinic.edges.size(); i += 2) {
        int u = dinic.edges[i].u;
        int v = dinic.edges[i].v;
        if (in_source_set[u] && !in_source_set[v]) {
            cout << u << " -> " << v << " (Capacity: " <<
dinic.edges[i].cap << ")\n";
        }
    }

    return 0;
}
*/

```

3.9 Min Cost Max Flow

```

/*
MIN COST MAX FLOW (Dijkstra with Johnson's Potentials)

Idea:
- Uses Dijkstra instead of SPFA to find the cheapest
augmenting path.
- Maintains a potential function 'pi' for each node to
handle negative
weight residual edges.
- Edge weights are reweighted:  $w'(u, v) = w(u, v) + pi[u] - pi[v] \geq 0$ .
- Complexity:  $O(F * E \log V)$  where  $F$  is the flow. Fast and
safe!
*/
const long long LINF = 1e18;

struct MCMFEdge {
    int u, v;
    long long cap, flow, cost;
};

```

```

};

struct MinCostMaxFlow {
    int n;
    vector<MCMFEdge> edges;
    vector<vector<int>>> adj;
    vector<long long> pi, dist;
    vector<int> parent_edge;

    MinCostMaxFlow(int _n) {
        n = _n;
        adj.assign(n + 1, vector<int>());
        pi.assign(n + 1, 0);
        dist.assign(n + 1, 0);
        parent_edge.assign(n + 1, -1);
    }

    void add_edge(int u, int v, long long cap, long long cost,
bool is_directed = true) {
        adj[u].push_back(edges.size());
        edges.push_back({u, v, cap, 0, cost});

        adj[v].push_back(edges.size());
        edges.push_back({v, u, is_directed ? 0 : cap, 0, -cost});
    }

    // Run SPFA once to set initial potentials.
    // ONLY UNCOMMENT inside mcmf() if the original graph has
NEGATIVE edge costs.
    void set_pi(int s) {
        fill(pi.begin(), pi.end(), LINF);
        vector<bool> in_queue(n + 1, false);
        queue<int> q;

        pi[s] = 0;
        q.push(s);
        in_queue[s] = true;

        while (!q.empty()) {
            int u = q.front();
            q.pop();
            in_queue[u] = false;

            for (int id : adj[u]) {
                int v = edges[id].v;
                if (edges[id].cap - edges[id].flow > 0 && pi[v] >
pi[u] + edges[id].cost) {
                    pi[v] = pi[u] + edges[id].cost;
                    if (!in_queue[v]) {
                        q.push(v);
                        in_queue[v] = true;
                    }
                }
            }
        }
    }

    // Dijkstra using reduced costs
    bool dijkstra(int s, int t) {
        fill(dist.begin(), dist.end(), LINF);
        fill(parent_edge.begin(), parent_edge.end(), -1);

        priority_queue<pair<long long, int>, vector<pair<long
long, int>>, greater<pair<long long, int>>> pq;

        dist[s] = 0;
        pq.push({0, s});

        while (!pq.empty()) {
            long long d = pq.top().first;
            int u = pq.top().second;
            pq.pop();

            if (d > dist[u]) continue;

            for (int id : adj[u]) {
                int v = edges[id].v;
                long long remain_cap = edges[id].cap - edges[id].flow;

                if (remain_cap > 0) {
                    long long reduced_cost = edges[id].cost + pi[u] -
pi[v];

```

```

        if (dist[v] > dist[u] + reduced_cost) {
            dist[v] = dist[u] + reduced_cost;
            parent_edge[v] = id;
            pq.push({dist[v], v});
        }
    }
}
return dist[t] != LINF;
}

pair<long long, long long> mcmf(int s, int t, long long
max_required_flow = LINF) {
    long long total_flow = 0;
    long long total_cost = 0;

    // Uncomment this line ONLY IF the initial graph contains
    // negative costs
    // set_pi(s);

    while (total_flow < max_required_flow && dijkstra(s, t)) {
        long long push = max_required_flow - total_flow;
        int cur = t;
        while (cur != s) {
            int id = parent_edge[cur];
            push = min(push, edges[id].cap - edges[id].flow);
            cur = edges[id].u;
        }

        cur = t;
        while (cur != s) {
            int id = parent_edge[cur];
            edges[id].flow += push;
            edges[id ^ 1].flow -= push;
            total_cost += push * edges[id].cost;
            cur = edges[id].u;
        }

        total_flow += push;

        // Update potentials for the next iteration
        for (int i = 1; i <= n; i++) {
            if (dist[i] != LINF) {
                pi[i] += dist[i];
            }
        }
    }

    return {total_flow, total_cost};
}

//
// PATH TRACING: Prints all edges that are actively carrying
// flow
//
void print_flow_paths() {
    cout << "Edges with positive flow (Min Cost Max Flow):\n";
    // Iterate by 2 to check only the original forward edges
    for (int i = 0; i < edges.size(); i += 2) {
        if (edges[i].flow > 0) {
            cout << edges[i].u << " -> " << edges[i].v
                << " | Flow: " << edges[i].flow << " / " <<
                edges[i].cap
                << " | Unit Cost: " << edges[i].cost
                << " | Total Edge Cost: " << edges[i].flow *
                edges[i].cost << "\n";
        }
    }
}
};

```

3.10 Dsu On Tree(Sack)

```

/*
DSU ON TREE (SACK)

Idea:
- Efficiently answers OFFLINE queries about subtrees in O(N
log N) time.

```

- Solves problems like: "For each node U, how many nodes in its subtree have a specific property?"

Algorithm:

1. Find the "heavy" child (largest subtree) for each node.
2. DFS traversal:
 - Traverse all "light" children, answer their queries, then CLEAR their data.
 - Traverse the "heavy" child, answer its queries, and KEEP its data.
 - Add the data of the light children and the current node itself.
 - Answer queries for the current node.
 - If the current node is a light child ('keep == false'), CLEAR all its data.

*/

```

const int N = 2e5 + 5;
vector<int> G[N];

```

```

// Variables for Euler tour and Heavy-Light logic
int in[N], out[N], rn[N], timer;
int sz[N], big[N];

```

// 1. First DFS: Precompute subtree sizes, Euler tour, and heavy children

```

void dfs_sz(int u, int par) {
    in[u] = ++timer;
    rn[timer] = u;
    sz[u] = 1;
    big[u] = 0;

    for (int v : G[u]) {
        if (v != par) {
            dfs_sz(v, u);
            sz[u] += sz[v];
            if (sz[big[u]] < sz[v]) {
                big[u] = v; // Update heavy child
            }
        }
    }
    out[u] = timer;
}

```

// Struct to manage the global state (Modify this according to the problem!)

// Example: Counting frequencies of values

```

struct GlobalState {
    int freq[N];
    int count_of_freq[N];
}

```

```

void add(int node) {
    // int val = a[node];
    // count_of_freq[++freq[val]]++;
}

```

```

void remove(int node) {
    // int val = a[node];
    // count_of_freq[freq[val]--]--;
}

```

```

// int get_answer(...) { return ...; }
} state;

```

// 2. Second DFS: DSU on Tree (Sack)

```

void dsu_on_tree(int u, int par, bool keep) {
    // Step 2.1: Traverse light children and DO NOT keep their
    // data
    for (int v : G[u]) {
        if (v != par && v != big[u]) {
            dsu_on_tree(v, u, false);
        }
    }
}

```

```

// Step 2.2: Traverse the heavy child and KEEP its data
if (big[u] != 0) {
    dsu_on_tree(big[u], u, true);
}

```

```
// Step 2.3: Add light children's data and the current
// node's data
for (int v : G[u]) {
    if (v != par && v != big[u]) {
        for (int i = in[v]; i <= out[v]; i++) {
            state.add(rn[i]);
        }
    }
}
state.add(u); // Add the current node itself

// Step 2.4: ANSWER QUERIES for node `u` here
// for (auto query : queries[u]) {
//     ans[query.id] = state.get_answer(query.k);
// }

// Step 2.5: If we are a light child, clear our data so it
// doesn't affect
// other branches of our parent.
if (!keep) {
    for (int i = in[u]; i <= out[u]; i++) {
        state.remove(rn[i]);
    }
}

/*
USAGE EXAMPLE:
signed main() {
    // ... Read graph and queries ...
    // ... Save queries as `queries[u].push_back({k,
    query_id})` ...
    timer = 0;
    dfs_sz(1, -1);
    // Start DSU on tree from the root, keeping is irrelevant
    // for the root
    dsu_on_tree(1, -1, false);
    // Print answers
    // ...
    return 0;
}
*/
```

3.11 MO On Tree

```
/*
MO'S ALGORITHM ON TREES (WITH UPDATES - 3D MO)
Idea:
- Flattens the tree into an array using Euler Tour. Each
node appears
  exactly TWICE: when we enter it (tin) and when we leave
  it (tout).
- A node is considered "in the current path" if it appears
  exactly ONCE
  in our active range [L, R]. We maintain a boolean array
  `f[u]` to
  toggle the presence of node `u`.

Query Translation (Path from U to V, assuming tin[U] <
tin[V]):
- Case 1: U is an ancestor of V.
  Query range: [tin[U], tin[V]]. LCA is already
  included.
- Case 2: U is NOT an ancestor of V.
  Query range: [tout[U], tin[V]]. LCA is NOT in
  this range,
  so we must manually add/remove the LCA during the
  query.

Complexity:
- Without updates (2D Mo): Block size = N / sqrt(Q), Time =
  O(N * sqrt(Q))
- With updates (3D Mo): Block size = N^(2/3), Time =
  O(N^(5/3))
*/
const int N = 2e5 + 5;
// For 3D Mo (with updates), BLOCK size should be roughly
N^(2/3).
// For N = 100,000, N^(2/3) is approx 2150.
const int BLOCK = 2150;

int n, q;
```

```
int a[N];
vector<int> G[N];

// Euler Tour & LCA
int tin[N], tout[N], timer;
int tour[N * 2]; // Needs to be 2 * N because each node
appears twice
int up[18][N], depth[N];

// Mo's State
int freq[N], distinct_count;
int f[N]; // f[x] = 1 if node x is currently in the path, 0
otherwise
int ans[N];

struct Query {
    int l, r, lca_idx, t, id;

    bool operator<(const Query& other) const {
        int bl = l / BLOCK;
        int br = r / BLOCK;
        int obl = other.l / BLOCK;
        int obr = other.r / BLOCK;
        if (bl != obl) return bl < obl;
        if (br != obr) return (bl & 1) ? br < obr : br > obr;
        return (br & 1) ? t < other.t : t > other.t;
    }
};

struct Update {
    int x, new_val, old_val;
};

vector<Query> queries;
vector<Update> updates;

// 1. Build Euler Tour and LCA table
void dfs(int u, int par) {
    up[0][u] = par;
    for (int i = 1; i < 18; i++) {
        if (up[i - 1][u] != -1) {
            up[i][u] = up[i - 1][up[i - 1][u]];
        } else {
            up[i][u] = -1;
        }
    }

    tin[u] = ++timer;
    tour[timer] = u;

    for (int v : G[u]) {
        if (v != par) {
            depth[v] = depth[u] + 1;
            dfs(v, u);
        }
    }

    tout[u] = ++timer;
    tour[timer] = u;
}

int get_lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    int delta = depth[u] - depth[v];
    for (int i = 0; i < 18; i++) {
        if ((delta >> i) & 1) u = up[i][u];
    }
    if (u == v) return u;
    for (int i = 17; i >= 0; i--) {
        if (up[i][u] != up[i][v]) {
            u = up[i][u];
            v = up[i][v];
        }
    }
    return up[0][u];
}

// 2. State Manipulation Functions
void add(int val) {
    if (++freq[val] == 1) distinct_count++;
}
```

```

void sub(int val) {
    if (freq[val]-- == 1) distinct_count--;
}

// Toggle a node: if it's already in the path, remove it.
Otherwise, add it.
void toggle(int idx) {
    int x = tour[idx];
    f[x] ^= 1;
    if (f[x]) add(a[x]);
    else sub(a[x]);
}

// Apply point update dynamically
void apply_update(int x, int val, int old_val) {
    if (f[x]) {
        add(val);
        sub(old_val);
    }
    a[x] = val;
}

/*
USAGE EXAMPLE:

signed main() {
    // ... Read N, Q, array `a`, and tree edges ...

    timer = 0;
    memset(up, -1, sizeof(up));
    depth[1] = 0;
    dfs(1, -1);

    // Note: If values are large, do Coordinate Compression
    on `a` and updates first!

    // ... Reading Queries and Updates ...
    int time_counter = 0;
    for (int i = 1; i <= q; i++) {
        int type; cin >> type;
        if (type == 1) { // Path Query
            int u, v; cin >> u >> v;
            if (tin[u] > tin[v]) swap(u, v); // Ensure tin[u] <=
            tin[v]

            int lca = get_lca(u, v);
            if (lca == u) {
                // u is ancestor of v -> LCA is already inside
                [tin[u], tin[v]]
                queries.push_back({tin[u], tin[v], -1,
                time_counter, (int)queries.size() + 1});
            } else {
                // u is NOT ancestor of v -> LCA is missing, need
                to add it manually
                queries.push_back({tout[u], tin[v], tin[lca],
                time_counter, (int)queries.size() + 1});
            }
        } else { // Point Update
            int x, val; cin >> x >> val;
            time_counter++;
            updates.push_back({x, val, a[x]});
            a[x] = val; // temporarily apply to get `old_val` for
            future queries
        }
    }

    // Rollback the array `a` to its initial state before
    processing Mo
    for (int i = (int)updates.size() - 1; i >= 0; i--) {
        a[updates[i].x] = updates[i].old_val;
    }

    sort(queries.begin(), queries.end());

    int L = 1, R = 0, T = 0;
    for (const auto& q : queries) {
        while (R < q.r) toggle(++R);
        while (L > q.l) toggle(--L);
        while (R > q.r) toggle(R--);
        while (L < q.l) toggle(L++);
    }
}

```

```

while (T < q.t) {
    apply_update(updates[T].x, updates[T].new_val,
    updates[T].old_val);
    T++;
}
while (T > q.t) {
    T--;
    apply_update(updates[T].x, updates[T].old_val,
    updates[T].new_val);
}

// Manually toggle LCA if needed
if (q.lca_idx != -1) toggle(q.lca_idx);

ans[q.id] = distinct_count;

// Untoggle LCA to restore state
if (q.lca_idx != -1) toggle(q.lca_idx);
}

for (int i = 1; i <= queries.size(); i++) {
    cout << ans[i] << "\n";
}
return 0;
}
*/

```

3.12 Knapsack On Tree

```

/*
DP KNAPSACK ON TREE (Prerequisite DP / Connected Component
Knapsack)
Idea:
- Problem: Choose a subset of nodes to maximize total value
such that the
    total weight <= W.
CONSTRAINT: If a node is chosen, its parent MUST also be
chosen.
(e.g., Course selection with prerequisites, or picking a
connected
component rooted at node 1).
- Naive DP merges subtrees taking  $O(N * W^2)$ .
OPTIMIZATION (DFS Flattening Trick):
- We flatten the tree using DFS Pre-Order traversal.
- For a node at pre-order index `i`, its entire subtree
spans contiguously
    from index `i` to `i + sz[u] - 1`.
- Transitions:
    1. Skip node `i`: We CANNOT pick any node in its subtree.
    We jump
        directly to index `i + sz[u]`.
    2. Take node `i`: We pay `weight[u]`, gain `value[u]`,
    and move to
        index `i + 1` (allowing us to pick its children).
Complexity:
- Time:  $O(N * W)$ 
- Space:  $O(N * W)$ 
*/
struct TreeKnapsack {
    int n;
    vector<vector<int>> adj;
    vector<long long> weight, value;

    // DFS state
    vector<int> sz;
    vector<int> rev_dfn; // rev_dfn[i] = node ID at DFS
    pre-order index 'i'
    int timer;

    TreeKnapsack(int _n) {
        n = _n;
        adj.assign(n + 1, vector<int>());
        weight.assign(n + 1, 0);
        value.assign(n + 1, 0);
        sz.assign(n + 1, 0);
        rev_dfn.assign(n + 1, 0);
        timer = 0;
    }

    // Add directed edge from parent to child
    void add_edge(int u, int v) {

```

```

    adj[u].push_back(v);
}

// Build the Pre-Order array and calculate subtree sizes
void dfs(int u) {
    timer++;
    rev_dfn[timer] = u;
    sz[u] = 1;

    for (int v : adj[u]) {
        dfs(v);
        sz[u] += sz[v];
    }
}

// Solves the knapsack problem for a specific weight limit `W`
long long solve(int root, int W) {
    timer = 0;
    dfs(root); // Flatten the tree

    // dp[i][w] = max value considering nodes in pre-order
    // from index `i` to `n`,
    // having exactly `w` weight capacity available.
    // We use size (n + 2) to safely handle the jump to `n + 1`
    // when skipping the last nodes.
    vector<vector<long long>> dp(n + 2, vector<long long>(W + 1, 0));

    // Iterate backwards through the pre-order indices
    for (int i = n; i >= 1; i--) {
        int u = rev_dfn[i];
        int jump = i + sz[u]; // The index to jump to if we skip
        // subtree of `u`

        for (int w = 0; w <= W; w++) {
            // Option 1: Skip this node (and entirely skip its
            // subtree)
            long long skip = dp[jump][w];

            // Option 2: Take this node (if current capacity
            // allows)
            long long take = 0;
            if (w >= weight[u]) {
                take = dp[i + 1][w - weight[u]] + value[u];
            }

            dp[i][w] = max(skip, take);
        }
    }

    // The answer is the state starting at pre-order index 1
    // with full capacity W
    return dp[1][W];
}

/*
USAGE EXAMPLE:

signed main() {
    // Suppose we have 5 nodes (1 is the root)
    int n = 5;
    TreeKnapsack tk(n);

    // Values and weights for nodes 1 to 5
    tk.weight = {0, 1, 2, 3, 2, 1}; // 1-indexed (index 0 is
    // unused)
    tk.value = {0, 5, 10, 12, 4, 3};

    // Tree structure: 1 is parent of 2 and 3. 2 is parent of
    // 4 and 5.
    // Directed edges from parent to child!
    tk.add_edge(1, 2);
    tk.add_edge(1, 3);
    tk.add_edge(2, 4);
    tk.add_edge(2, 5);

    int weight_limit = 4;

    // Calculate max value if we root at 1 and have max
    weight 4

```

```

    long long max_val = tk.solve(1, weight_limit);

    cout << "Max value within weight " << weight_limit << "
    is: " << max_val << "\n";
    // Output expected: node 1 (w=1, v=5) + node 2 (w=2,
    // v=10) + node 5 (w=1, v=3)
    // Total weight = 4, Total value = 18.

    return 0;
}
*/

```

4 DP

4.1 DP Digit Optimization 1

```

/*
DIGIT DP OPTIMIZATION 1: REMOVING THE `is_less` (fr)
DIMENSION
*/

vector<int> R;
pair<ll, ll> dp[20];

pair<ll, ll> calc(int pos, int fr) {
    if (pos == -1) return make_pair(0, 1);
    auto &res = dp[pos];
    if (fr && res != make_pair(-1LL, -1LL)) return res;
    int lim_r = (fr ? 9 : R[pos]);
    pair<ll, ll> val = make_pair(0, 0);
    for (int i = 0; i <= lim_r; i++) {
        pair<ll, ll> nxt = calc(pos - 1, fr | (i < lim_r));
        val.first += nxt.first + i * nxt.second;
        val.second += nxt.second;
    }
    return (fr ? res = val : val);
}

```

4.2 DP Digit Optimization 2

```

/*
DIGIT DP OPTIMIZATION 2: ONE-TIME MEMSET FOR MULTI-TEST
CASES
*/

const int mod = 1e9 + 7;

string R;
int dp[51][51][51][51];

int calc(int pos, int c3, int c6, int c9, int fr) {
    if (pos == -1) {
        return (c3 > 0 && c3 == c6 && c6 == c9);
    }
    auto &res = dp[pos][c3][c6][c9];
    if (fr && res != -1) return res;
    ll val = 0;
    int lim_r = (fr ? 9 : R[pos] - '0');
    for (int i = 0; i <= lim_r; i++) {
        val += calc(pos - 1, c3 + (i == 3), c6 + (i == 6), c9 + (i
        == 9), fr | (i < lim_r));
        if (val >= mod) val -= mod;
    }
    return (fr ? res = val : val);
}

int solve(string x) {
    reverse(x.begin(), x.end());
    R = x;
    return calc((int)R.size() - 1, 0, 0, 0, 0);
}

bool check(string x) {
    int c3 = 0, c6 = 0, c9 = 0;
    for (auto i : x) {
        c3 += (i == '3');
        c6 += (i == '6');
        c9 += (i == '9');
    }
    return (c3 > 0 && c3 == c6 && c6 == c9);
}

```

```

}

/*
// Usage in main():
// memset(dp, -1, sizeof dp); // Call this ONLY ONCE
// while (test--) {
//     string a, b; cin >> a >> b;
//     cout << (1LL * solve(b) - solve(a) + check(a) + mod) %
mod << "\n";
// }
*/

```

4.3 DP SoS

```

const int LOG = 20;
int dp[1 << LOG];

// 1. Sum over Submasks: dp[mask] = \sum_{sub \subseteq mask} A[sub]
// (sub is formed by turning OFF some bits of mask)
void sos_submask(int n, int a[]) {
    for (int mask = 0; mask < (1 << n); mask++) {
        dp[mask] = a[mask];
    }
    for (int i = 0; i < n; i++) {
        for (int mask = 0; mask < (1 << n); mask++) {
            if ((mask >> i) & 1) {
                dp[mask] += dp[mask ^ (1 << i)];
            }
        }
    }
}

// 2. Sum over Supermasks: dp[mask] = \sum_{mask \subseteq sup} A[sup]
// (sup is formed by turning ON some bits of mask)
void sos_supermask(int n, int a[]) {
    for (int mask = 0; mask < (1 << n); mask++) {
        dp[mask] = a[mask];
    }
    for (int i = 0; i < n; i++) {
        for (int mask = 0; mask < (1 << n); mask++) {
            if (!((mask >> i) & 1)) {
                dp[mask] += dp[mask ^ (1 << i)];
            }
        }
    }
}

/*
COMMON USAGE PATTERNS & TRICKS:
1. Find pairs (u, v) such that u & v = 0:
2. Number of elements `x` such that (x & mask) == x:
3. Number of elements `x` such that (x & mask) == mask:
4. Finding two elements x, y with maximum (x | y):
*/

```

4.4 DP Divide And Conquer

```

/*
DIVIDE AND CONQUER DP OPTIMIZATION

Idea:
- Optimizes DP equations of the form:
  dp[i][j] = min_{k < j} (dp[i-1][k] + cost(k+1, j))
- Condition: The optimal split point `opt(i, j)` must be monotonic,
  meaning: opt(i, j) <= opt(i, j+1).
- This monotonicity allows us to use Divide & Conquer: if
  we know the optimal split point for mid is `opt_mid`, then the split
  point for any j < mid is <= opt_mid, and for any j > mid is >=
  opt_mid.

Complexity:
- Original DP: O(K * N^2)
- D&C Optimized: O(K * N log N)

Memory Optimization:

```

```

- We only need the DP state from the previous phase (i - 1), so we can
  use two 1D arrays: `dp_before` and `dp_current`.
*/
const long long LINF = 1e18;

// DP arrays (1-indexed for the items)
vector<long long> dp_before, dp_current;

// 1. Problem-specific cost function
// Note: To keep the overall complexity O(K * N log N), this
// cost function
// MUST be calculable in O(1). Usually done via prefix sums or
// sliding window.
long long cost(int l, int r) {
    if (l > r) return 0;
    // TODO: Replace with the actual O(1) cost logic of the
    // problem
    return 0;
}

// 2. Divide and Conquer recursive function
// l, r: the range of `j` we are currently computing
// dp_current[j] for.
// optl, opttr: the valid range of the optimal split point `k`
// for this range.
void compute(int l, int r, int optl, int opttr) {
    if (l > r) return;

    int mid = l + (r - l) / 2;

    // We want to find the optimal split point `k` for
    // dp_current[mid]
    pair<long long, int> best = {LINF, -1};

    // k is the end of the previous segment, so the new segment
    // is [k+1, mid].
    // Note: k must be strictly less than `mid` if we require
    // non-empty segments,
    // or k <= mid depending on the problem formulation.
    for (int k = optl; k <= min(mid - 1, opttr); k++) {
        // Check if the previous state is valid
        if (dp_before[k] != LINF) {
            long long current_val = dp_before[k] + cost(k + 1, mid);
            if (current_val < best.first) {
                best = {current_val, k};
            }
        }
    }

    dp_current[mid] = best.first;
    int opt = best.second;

    // 3. Recurse for the left and right halves using the
    // discovered monotonicity
    // For j < mid, optimal k is in [optl, opt]
    compute(l, mid - 1, optl, opt);

    // For j > mid, optimal k is in [opt, opttr]
    compute(mid + 1, r, opt, opttr);
}

```

```

/*
USAGE EXAMPLE:

signed main() {
    int n = 1000; // Number of elements
    int K = 10;   // Number of partitions

    dp_before.assign(n + 1, LINF);
    dp_current.assign(n + 1, LINF);

    // Base case for 0 partitions
    dp_before[0] = 0;
    // dp_before[x > 0] is LINF because we can't partition x
    // items into 0 parts

    for (int i = 1; i <= K; i++) {
        // Reset current DP array for the new partition step
        fill(dp_current.begin(), dp_current.end(), LINF);
    }
}

```

```

    // Compute dp_current[1...n], knowing optimal k is in
    [0...n-1]
    compute(1, n, 0, n - 1);

    // Move current row to before row for the next
    iteration
    dp_before = dp_current;
}

cout << "Min cost to partition " << n << " items into "
<< K
    << " parts is: " << dp_current[n] << "\n";

return 0;
}
*/

```

5 Strings

5.1 KMP

```

vector<int> compute_pi(const string& p) {
    int m = p.length();
    vector<int> pi(m, 0);
    for (int i = 1, j = 0; i < m; i++) {
        while (j > 0 && p[i] != p[j]) {
            j = pi[j - 1];
        }
        if (p[i] == p[j]) {
            j++;
        }
        pi[i] = j;
    }
    return pi;
}

```

5.2 Z Function

```

vector<int> z_function(const string& s) {
    int n = (int)s.length();
    vector<int> z(n, 0);
    for (int i = 1, l = 0, r = 0; i < n; i++) {
        if (i <= r) {
            z[i] = min(r - i + 1, z[i - l]);
        }
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if (i + z[i] - 1 > r) {
            l = i;
            r = i + z[i] - 1;
        }
    }
    return z;
}

```

5.3 Suffix Array

```

/*
    SUFFIX ARRAY & LCP ARRAY (O(N log N) Construction)
    Idea:
    - Suffix Array (sa): Sorts all suffixes of a string
    lexicographically.
    sa[i] stores the starting index of the i-th
    lexicographically
    smallest suffix.
    - LCP Array (lcp): Stores the Longest Common Prefix between
    consecutive
    suffixes in the sorted Suffix Array. lcp[i] = LCP(sa[i],
    sa[i-1]).
    Algorithm:
    - Appends a special character '$' (strictly smaller than
    all other characters)
    to the end of the string.
    - Sorts substrings of length 2^k using Radix Sort in O(N).
    - Repeats log(N) times -> Total time O(N log N).
    - Computes LCP in O(N) using Kasai's Algorithm.
*/

```

```

struct SuffixArray {
    int n;
    string s;
    vector<int> sa; // Suffix Array
    vector<int> c; // Equivalence classes
    vector<int> lcp; // Longest Common Prefix array

    // Constructor automatically builds SA and LCP
    SuffixArray(string const& _s) : s(_s + "$") {
        n = s.length();
        sa.assign(n, 0);
        c.assign(n, 0);
        lcp.assign(n, 0);

        build_sa();
        build_lcp();
    }
}

```

```

void build_sa() {
    const int alphabet = 256;
    vector<int> cnt(max(alphabet, n), 0);
    vector<int> p(n), pn(n), cn(n);

    // 1. Initial sort for length 1 (k = 0)
    for (int i = 0; i < n; i++) cnt[s[i]]++;
    for (int i = 1; i < alphabet; i++) cnt[i] += cnt[i - 1];
    for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;

    c[p[0]] = 0;
    int classes = 1;
    for (int i = 1; i < n; i++) {
        if (s[p[i]] != s[p[i - 1]]) classes++;
        c[p[i]] = classes - 1;
    }

    // 2. Transitions for length 2^h
    for (int h = 0; (1 << h) < n; h++) {
        // Shift by 2^h
        for (int i = 0; i < n; i++) {
            pn[i] = p[i] - (1 << h);
            if (pn[i] < 0) pn[i] += n;
        }

        // Radix Sort on the first half
        fill(cnt.begin(), cnt.begin() + classes, 0);
        for (int i = 0; i < n; i++) cnt[c[pn[i]]]++;
        for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
        for (int i = n - 1; i >= 0; i--) p[--cnt[c[pn[i]]]] =
            pn[i];

        // Update equivalence classes
        cn[p[0]] = 0;
        classes = 1;
        for (int i = 1; i < n; i++) {
            pair<int, int> cur = {c[p[i]], c[(p[i] + (1 << h)) %
                n]};
            pair<int, int> prev = {c[p[i - 1]], c[(p[i - 1] + (1
                << h)) % n]};
            if (cur != prev) classes++;
            cn[p[i]] = classes - 1;
        }
        c.swap(cn);
    }
    sa = p;
}

// Kasai's algorithm to build LCP array in O(N)
void build_lcp() {
    vector<int> rank(n, 0);
    for (int i = 0; i < n; i++) {
        rank[sa[i]] = i;
    }

    int k = 0;
    // Iterate through original string indices (excluding the
    appended '$')
    for (int i = 0; i < n - 1; i++) {
        int pi = rank[i];
        int j = sa[pi - 1]; // The suffix preceding the current
        one in SA

        // Expand the matching prefix
    }
}

```

```

while (i + k < n && j + k < n && s[i + k] == s[j + k]) {
    k++;
}

lcp[pi] = k;
if (k > 0) k--; // Move to the next suffix, LCP
decreases by at most 1
}
}
};

/*
USAGE EXAMPLES:

signed main() {
    string str = "ababba";
    SuffixArray sa(str);

    // 1. Number of Distinct Substrings
    // Formula: Total substrings - sum of LCP array
    // Total substrings of length N is N * (N + 1) / 2
    long long N = str.length();
    long long distinct_substrings = N * (N + 1) / 2;
    for (int i = 1; i <= N; i++) {
        distinct_substrings -= sa.lcp[i];
    }
    cout << "Distinct substrings: " << distinct_substrings <<
    "\n";

    // 2. Printing Suffix Array & LCP Array
    cout << "i\tSA[i]\tLCP[i]\tSuffix\n";
    // Note: sa.sa[0] is always the appended '$' (length 0
    suffix)
    for (int i = 1; i <= N; i++) {
        cout << i << "\t"
            << sa.sa[i] << "\t"
            << sa.lcp[i] << "\t"
            << str.substr(sa.sa[i]) << "\n";
    }

    return 0;
}
*/

```

5.4 Aho Corasick

```

/*
AHO-CORASICK AUTOMATON
Idea:
- Multi-pattern string search algorithm.
- Constructs a Trie from a set of patterns and adds:
    1. Failure Links (`fail`): Where to go when a character
    mismatch occurs.
    2. Output Links (`output`): Direct link to the nearest
    valid pattern
        ending in the failure chain (handles
        overlapping/substring patterns).
- OPTIMIZATION: We convert the Trie into a Deterministic
Finite Automaton
(DFA). Transition `next[c]` either points to the actual
child OR the
fallback state in O(1).

Complexity:
- Build Time: O(Sum of pattern lengths * Alphabet Size)
- Search Time: O(Length of Text + Number of Matches)
*/

const int ALPHABET = 26;
const char BASE_CHAR = 'a'; // Change to 'A' or '0' depending
on the problem

struct ACNode {
    int next[ALPHABET];
    int fail;
    int output; // Also known as dict_link (points to the
nearest terminal node)
    vector<int> pattern_ids; // IDs of patterns that end exactly
at this node

ACNode() {

```

```

fill(begin(next), end(next), 0);
fail = 0;
output = 0;
}
};

struct AhoCorasick {
    vector<ACNode> t;

    AhoCorasick() {
        t.emplace_back(); // Root node at index 0
    }

    // 1. Insert a pattern into the Trie
    void insert(const string& s, int id) {
        int cur = 0;
        for (char c : s) {
            int v = c - BASE_CHAR;
            if (!t[cur].next[v]) {
                t[cur].next[v] = t.size();
                t.emplace_back();
            }
            cur = t[cur].next[v];
        }
        t[cur].pattern_ids.push_back(id);
    }

    // 2. Build Failure and Output links via BFS
    void build() {
        queue<int> q;

        // Initialize root's children
        for (int i = 0; i < ALPHABET; i++) {
            if (t[0].next[i]) {
                t[t[0].next[i]].fail = 0;
                q.push(t[0].next[i]);
            }
        }

        while (!q.empty()) {
            int u = q.front();
            q.pop();

            for (int i = 0; i < ALPHABET; i++) {
                int v = t[u].next[i];
                if (v) {
                    // If child exists, set its failure link to the
child of u's fail node
                    t[v].fail = t[t[u].fail].next[i];

                    // Optimization: Output link points to the nearest
pattern end
                    if (!t[t[v].fail].pattern_ids.empty()) {
                        t[v].output = t[t[v].fail];
                    } else {
                        t[v].output = t[t[v].fail].output;
                    }

                    q.push(v);
                } else {
                    // Trie Automaton Optimization: If child doesn't
exist,
                    // point directly to the fallback state. This makes
text traversal O(1).
                    t[u].next[i] = t[t[u].fail].next[i];
                }
            }
        }
    }

    // 3. Search for patterns in the given text
    void search(const string& text) {
        int cur = 0;
        for (int i = 0; i < text.length(); i++) {
            int c = text[i] - BASE_CHAR;
            cur = t[cur].next[c]; // O(1) transition due to
Automaton optimization

            // Traverse output links to find all patterns ending at
current position i
            int temp = cur;

```

```

while (temp != 0) {
    for (int id : t[temp].pattern_ids) {
        // You can store or process the match here
        cout << "Pattern ID " << id << " ends at index " <<
            i << "\n";
    }
    // Jump to the next valid pattern in the failure chain
    // If the current node doesn't have an end word but
    // has an output link,
    // this safely jumps to it.
    if (t[temp].pattern_ids.empty()) {
        temp = t[temp].output;
    } else {
        temp = t[temp].output;
    }
}
}
};

/*
USAGE EXAMPLE:

signed main() {
    AhoCorasick ac;

    vector<string> patterns = {"he", "she", "his", "hers"};
    for (int i = 0; i < patterns.size(); i++) {
        ac.insert(patterns[i], i);
    }

    // MUST call build() after inserting all patterns and
    // before searching
    ac.build();

    string text = "ushers";
    cout << "Searching in text: " << text << "\n";
    ac.search(text);

    // Output expected:
    // Pattern ID 1 ("she") ends at index 3
    // Pattern ID 0 ("he") ends at index 3
    // Pattern ID 3 ("hers") ends at index 5

    return 0;
}
*/

```

6 Math

6.1 Extended Euclid

```

// 1. Core Extended GCD
// Returns gcd(a, b) and updates x, y via reference.
long long extgcd(long long a, long long b, long long &x, long
long &y) {
    if (b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extgcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

// 2. Solve Linear Diophantine Equation: ax + by = c
// Returns true if a solution exists, false otherwise.
// If true, (x0, y0) stores one valid base solution, and g
stores gcd(a, b).
//
// GENERAL SOLUTION GENERATOR:
// All solutions can be generated using an integer k:
// x = x0 + k * (b / g)
// y = y0 - k * (a / g)
bool solve_diophantine(long long a, long long b, long long c,
long long &x0, long long &y0, long long &g) {
    // Solve for abs(a) and abs(b) first to maintain correct
    signs

```

```

g = extgcd(abs(a), abs(b), x0, y0);

if (c % g != 0) {
    return false; // No solution if c is not divisible by
gcd(a, b)
}

x0 *= c / g;
y0 *= c / g;

// Adjust signs back
if (a < 0) x0 = -x0;
if (b < 0) y0 = -y0;

return true;
}

// 3. Modular Multiplicative Inverse
// Solves: ax = 1 (mod m).
// Valid ONLY IF a and m are coprime (gcd(a, m) == 1).
// Returns -1 if the inverse does not exist.
long long mod_inverse(long long a, long long m) {
    long long x, y;
    long long g = extgcd(a, m, x, y);

    if (g != 1) {
        return -1; // Inverse doesn't exist
    }

    // x can be negative, so we adjust it to be in the range [0,
m - 1]
    return (x % m + m) % m;
}

```

6.2 CRT

```

/*
GENERALIZED CHINESE REMAINDER THEOREM (CRT)
Idea:
- Solves a system of linear congruences:
    x = a_1 (mod m_1)
    x = a_2 (mod m_2)
    ...
    x = a_n (mod m_n)
- Unlike the standard CRT, this GENERALIZED version does
NOT require
the moduli (m_i) to be pairwise coprime.
- It works by iteratively merging two congruences into one.
Math:
- Merging x = a1 (mod m1) and x = a2 (mod m2):
    We need x = a1 + k1 * m1 = a2 + k2 * m2
    => k1 * m1 - k2 * m2 = a2 - a1
    This is a linear Diophantine equation. Let g = gcd(m1,
m2).
    If (a2 - a1) is not divisible by g, there is NO SOLUTION.
    Otherwise, we solve for k1 and find the new merged
congruence:
    x = new_a (mod lcm(m1, m2)).
Complexity: O(N log(LCM))
Note: Uses __int128_t to prevent overflow during
intermediate multiplications.
*/
long long extgcd(long long a, long long b, long long &x, long
long &y) {
    if (b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extgcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

// Generalized CRT
// Inputs: vector of remainders `a`, vector of moduli `m`.
// Returns: A pair {ans_a, ans_m} such that x = ans_a (mod
ans_m).
// If no solution exists, returns {-1, -1}.

```

```

pair<long long, long long> crt(const vector<long long>& a,
    const vector<long long>& m) {
    assert(a.size() == m.size());
    if (a.empty()) return {0, 1}; // Trivial case

    long long ans_a = a[0];
    long long ans_m = m[0];

    for (int i = 1; i < (int)a.size(); i++) {
        long long cur_a = a[i];
        long long cur_m = m[i];

        long long u, v;
        long long g = extgcd(ans_m, cur_m, u, v);

        // Check if a solution exists for this pair
        if ((cur_a - ans_a) % g != 0) {
            return {-1, -1}; // No solution
        }

        long long lcm = (ans_m / g) * cur_m;

        // Calculate k1 = (a2 - a1) / g * u (mod m2 / g)
        // Using __int128_t to avoid overflow when multiplying
        // (cur_a - ans_a) / g and u
        long long diff = cur_a - ans_a;
        __int128_t k = (__int128_t)diff / g * u;
        k %= (cur_m / g);

        // Substitute k back to find the new remainder
        ans_a = ans_a + (long long)k * ans_m;

        // Normalize ans_a to be positive and within [0, lcm - 1]
        ans_a = (ans_a % lcm + lcm) % lcm;
        ans_m = lcm;
    }

    return {ans_a, ans_m};
}

/*
USAGE EXAMPLE:

signed main() {
    // System of equations:
    // x = 2 (mod 3)
    // x = 3 (mod 5)
    // x = 2 (mod 7)

    vector<long long> a = {2, 3, 2};
    vector<long long> m = {3, 5, 7};

    pair<long long, long long> res = crt(a, m);

    if (res.first == -1) {
        cout << "No solution exists.\n";
    } else {
        cout << "Smallest positive solution: x = " << res.first
            << "\n";
        cout << "General solution: x = " << res.first << " (mod
            " << res.second << ")\n";
    }

    // Output:
    // Smallest positive solution: x = 23
    // General solution: x = 23 (mod 105)

    return 0;
}
*/

```

6.3 Lucas Theorem

```

/*
LUCAS THEOREM
Idea:
- Computes  $nCr(n, k) \% p$  where  $n$  and  $k$  can be very large
  (up to  $10^{18}$ ),
  but  $p$  is a relatively small PRIME number (e.g.,  $p \leq 10^6$ ).
*/

```

```

- Theorem:  $nCr(n, k) = \text{Product of } nCr(n_i, k_i) \pmod p$ ,
  where  $n_i$  and  $k_i$  are the digits of  $n$  and  $k$  in base  $p$ .
- If at any point  $k_i > n_i$ , then  $nCr(n_i, k_i) = 0$ ,
  meaning the entire result becomes 0.
Complexity:
- Precomputation:  $O(p)$  to compute factorials up to  $p - 1$ .
- Query:  $O(\log_p(n))$  per  $nCr$  query.
- Space:  $O(p)$ 
Requirement:
-  $p$  MUST be a prime number.
*/
struct LucasTheorem {
    long long p;
    vector<long long> fact, invFact;

    long long power(long long base, long long exp) {
        long long res = 1;
        base %= p;
        while (exp > 0) {
            if (exp % 2 == 1) res = (res * base) % p;
            base = (base * base) % p;
            exp /= 2;
        }
        return res;
    }

    long long modInverse(long long n) {
        return power(n, p - 2);
    }

    // Constructor: Precomputes factorials and inverse
    // factorials up to  $p - 1$ 
    LucasTheorem(long long prime_mod) {
        p = prime_mod;
        fact.assign(p, 1);
        invFact.assign(p, 1);

        for (int i = 1; i < p; i++) {
            fact[i] = (fact[i - 1] * i) % p;
        }

        invFact[p - 1] = modInverse(fact[p - 1]);
        for (int i = p - 2; i >= 0; i--) {
            invFact[i] = (invFact[i + 1] * (i + 1)) % p;
        }

        // Computes  $nCr(n, k) \% p$  for small values ( $n, k < p$ )
        long long nCr_small(long long n, long long k) {
            if (k < 0 || k > n) return 0;
            long long num = fact[n];
            long long den = (invFact[k] * invFact[n - k]) % p;
            return (num * den) % p;
        }

        // Computes  $nCr(n, k) \% p$  for large values using Lucas
        // Theorem
        long long lucas(long long n, long long k) {
            if (k < 0 || k > n) return 0;

            long long res = 1;
            while (n > 0 || k > 0) {
                long long ni = n % p;
                long long ki = k % p;

                if (ki > ni) return 0;

                res = (res * nCr_small(ni, ki)) % p;

                n /= p;
                k /= p;
            }
            return res;
        }
    };
}

```

6.4 Euler Totient

```

/*

```

```

EULER'S TOTIENT FUNCTION (PHI FUNCTION)
Idea:
- phi(n) counts the number of integers between 1 and n
  inclusive
  that are coprime to n (i.e., gcd(x, n) == 1).
Important Properties for CP:
1. If p is prime: phi(p) = p - 1
2. If p is prime and k >= 1: phi(p^k) = p^k - p^(k-1)
3. Multiplicative: If gcd(a, b) == 1, then phi(a * b) =
  phi(a) * phi(b)
4. Euler's Theorem: If gcd(a, m) == 1, then a^phi(m) = 1
  (mod m)
5. Sum of phi over divisors: The sum of phi(d) for all
  divisors d of n
  is exactly equal to n. (Sum_{d|n} phi(d) = n)
*/
// 1. Single Computation
// Computes phi(n) in O(sqrt(n)) time.
// Uses the formula: phi(n) = n * (1 - 1/p1) * (1 - 1/p2) *
...
// where p_i are the distinct prime factors of n.
long long get_phi(long long n) {
    long long result = n;
    for (long long i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0) {
                n /= i;
            }
            result -= result / i;
        }
    }
    if (n > 1) {
        result -= result / n;
    }
    return result;
}

// 2. Sieve Computation (Batch Computation)
// Computes phi(i) for all 1 <= i <= N in O(N log log N) time.
// Similar to the Sieve of Eratosthenes.
vector<int> phi_sieve(int n) {
    vector<int> phi(n + 1);

    for (int i = 0; i <= n; i++) {
        phi[i] = i;
    }

    for (int i = 2; i <= n; i++) {
        if (phi[i] == i) {
            for (int j = i; j <= n; j += i) {
                phi[j] -= phi[j] / i;
            }
        }
    }
    return phi;
}

/*
USAGE EXAMPLE:
signed main() {
    // 1. Single query for a large number
    long long n = 12;
    cout << "phi(" << n << ") = " << get_phi(n) << "\n";
    // Output: 4 (Coprimes are 1, 5, 7, 11)
    // 2. Precomputing phi for all numbers up to MAX_N
    int MAX_N = 100000;
    vector<int> phi = phi_sieve(MAX_N);
    cout << "phi(10) = " << phi[10] << "\n"; // Output: 4
    cout << "phi(97) = " << phi[97] << "\n"; // Output: 96
    (Since 97 is prime)
    return 0;
}
*/

```

6.5 Mobius Function

```

/*
MOBIUS FUNCTION & INVERSION

Definition of mu(n):
- mu(n) = 1 if n = 1

```

```

- mu(n) = 0 if n has a squared prime factor (e.g., 4, 9,
  12, 18)
- mu(n) = 1 if n is square-free with an EVEN number of
  prime factors
- mu(n) = -1 if n is square-free with an ODD number of
  prime factors

```

Important Properties for CP:

- Sum over divisors: The sum of mu(d) for all divisors d of n is 1 if n = 1, and 0 otherwise. (Sum_{d|n} mu(d) = [n == 1])
-> This is the most used property to replace [gcd(i, j) == 1].
- Mobius Inversion Formula:
If g(n) = Sum_{d|n} f(d), then f(n) = Sum_{d|n} mu(d) * g(n/d).

```

*/
// 1. Single Computation
// Computes mu(n) in O(sqrt(n)) time.
int get_mu(long long n) {
    if (n == 1) return 1;
    int prime_count = 0;

    for (long long i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            int count = 0;
            while (n % i == 0) {
                count++;
                n /= i;
            }
            if (count > 1) return 0;
            prime_count++;
        }
    }
    if (n > 1) {
        prime_count++;
    }

    return (prime_count % 2 == 1) ? -1 : 1;
}

// 2. Linear Sieve Computation (Batch Computation)
// Computes mu(i) for all 1 <= i <= N in exactly O(N) time.
struct MobiusSieve {
    int n;
    vector<int> mu;
    vector<int> primes;
    vector<bool> is_prime;

    MobiusSieve(int _n) {
        n = _n;
        mu.assign(n + 1, 0);
        is_prime.assign(n + 1, true);

        is_prime[0] = is_prime[1] = false;
        mu[1] = 1;

        for (int i = 2; i <= n; i++) {
            if (is_prime[i]) {
                primes.push_back(i);
                mu[i] = -1; // Prime numbers have exactly 1 prime
                           factor (ODD)
            }
            for (int p : primes) {
                if (i * p > n) break;
                is_prime[i * p] = false;

                if (i % p == 0) {
                    // i * p is divisible by p^2
                    mu[i * p] = 0;
                    break;
                } else {
                    // p is a new distinct prime factor
                    mu[i * p] = -mu[i];
                }
            }
        }
    }
};
/*

```

USAGE EXAMPLE:

```
signed main() {
    // 1. Single query for a large number
    long long n = 30; // 30 = 2 * 3 * 5 (3 prime factors ->
    odd -> mu = -1)
    cout << "mu(" << n << ") = " << get_mu(n) << "\n"; //
    Output: -1

    long long m = 12; // 12 = 2^2 * 3 (has squared prime
    factor 2^2 -> mu = 0)
    cout << "mu(" << m << ") = " << get_mu(m) << "\n"; //
    Output: 0

    // 2. Precomputing mu for all numbers up to MAX_N in O(N)
    int MAX_N = 100000;
    MobiusSieve sieve(MAX_N);

    cout << "mu(10) = " << sieve.mu[10] << "\n"; // 10 = 2 *
    5 -> Output: 1
    cout << "mu(97) = " << sieve.mu[97] << "\n"; // 97 is
    prime -> Output: -1

    return 0;
}
*/
```

6.6 Miller Rabin

```
/*
    MILLER-RABIN PRIMALITY TEST

    Idea:
    - A fast primality test. While originally probabilistic, it
    can be made
    DETERMINISTIC for all 64-bit integers by testing a
    specific, small set
    of prime bases.
    - To test if n is prime, we write  $n - 1 = d * 2^s$ , where d
    is odd.
    - For a base 'a', n is composite if  $a^d \not\equiv 1 \pmod{n}$  AND
     $a^{(d * 2^r)} \not\equiv n - 1 \pmod{n}$ 
    for all r in [0, s - 1].

    Bases for Deterministic Test:
    - For  $n < 2^{64}$ , testing the first 12 primes is strictly
    sufficient:
    {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}

    Complexity:
    - Time:  $O(K \log n)$  where K is the number of bases ( $K \leq$ 
    12).
    - Space:  $O(1)$ 
    - Uses __uint128_t to cleanly prevent overflow during
    64-bit multiplications.
*/
using u64 = uint64_t;
using u128 = __uint128_t;

// Modular multiplication using 128-bit integer to avoid
// overflow
u64 mod_mul(u64 a, u64 b, u64 m) {
    return (u64)((u128)a * b % m);
}

// Modular exponentiation: (base^exp) % mod
u64 mod_pow(u64 base, u64 exp, u64 mod) {
    u64 res = 1;
    base %= mod;
    while (exp > 0) {
        if (exp % 2 == 1) res = mod_mul(res, base, mod);
        base = mod_mul(base, base, mod);
        exp /= 2;
    }
    return res;
}

// Miller-Rabin deterministic primality test for 64-bit
// integers
bool is_prime(u64 n) {
    if (n < 2) return false;
```

```
if (n == 2 || n == 3) return true;
if (n % 2 == 0) return false;

// Write n - 1 as d * 2^s
u64 d = n - 1;
int s = 0;
while (d % 2 == 0) {
    d /= 2;
    s++;
}

// Deterministic bases for values up to 2^64
static const u64 bases[] = {2, 3, 5, 7, 11, 13, 17, 19, 23,
29, 31, 37};

for (u64 a : bases) {
    // If n is smaller than or equal to the base, we are done
    // (since n is prime and has already passed the n % 2 == 0
    check,
    // it would just be one of the prime bases).
    if (n <= a) break;

    u64 x = mod_pow(a, d, n);
    if (x == 1 || x == n - 1) continue;

    bool composite = true;
    for (int r = 1; r < s; r++) {
        x = mod_mul(x, x, n);
        if (x == n - 1) {
            composite = false;
            break;
        }
    }

    if (composite) return false;
}

return true;
}

/*
    USAGE EXAMPLE:

    signed main() {
        long long n = 1000000000000000000000000LL; // A large 64-bit
        number

        if (is_prime(n)) {
            cout << n << " is PRIME.\n";
        } else {
            cout << n << " is COMPOSITE.\n";
        }

        // Expected Output: 1000000000000000000000000 is PRIME.

        return 0;
    }
*/
```

7 Geometry

8 Miscellaneous

8.1 Alien Trick

```
/*
    ALIEN'S TRICK (WQS BINARY SEARCH / LAGRANGE RELAXATION)

    Idea:
    - Solves DP problems with an "exactly K items/partitions"
    constraint.
    - We drop the K constraint and add a penalty `lambda` for
    each partition.
    - We binary search on `lambda`. If we use > K partitions,
    we increase
    the penalty. If we use < K partitions, we decrease it.
    - To handle collinear points on the convex hull, we always
    prefer the
    MAXIMUM (or MINIMUM) number of partitions for the same
    cost, and use
```

```

    `cnt` >= K` in the binary search.

Condition:
- Let  $f(x)$  be the optimal cost using exactly  $x$  partitions.
-  $f(x)$  MUST be convex or concave.

Complexity:  $O(N * \log(\text{MAX\_PENALTY}))$  instead of  $O(N * K)$ .
*/
const long long LINF = 1e18;

int n;
vector<long long> a; // Input array

// 1. The 1D DP function without the 'K' constraint
// Returns a pair: {optimal_cost_with_penalty,
// number_of_partitions_used}
pair<long long, int> solve_dp(long long lambda) {
    // dp[i] stores {min_cost, max_partitions_used} up to index
    // i
    vector<pair<long long, int>> dp(n + 1, {LINF, 0});
    dp[0] = {0, 0};

    for (int i = 1; i <= n; i++) {
        // Iterate over all possible previous split points
        // Note: In real problems, this inner loop is often
        // optimized to  $O(1)$ 
        // using Convex Hull Trick, D&C, or Monotone Queue.
        for (int j = 0; j < i; j++) {
            // cost(j+1, i) is the cost of the segment from j+1 to i
            // We ADD `lambda` as a penalty for creating this new
            // segment
            long long segment_cost = 0; /* TODO: Calculate real cost */
            long long new_cost = dp[j].first + segment_cost +
                lambda;
            int new_cnt = dp[j].second + 1;

            // We want to MINIMIZE cost.
            // TIE-BREAKER: If costs are equal, MAXIMIZE the number
            // of partitions.
            // This tie-breaker is CRUCIAL to handle collinear
            // points on the convex hull!
            if (new_cost < dp[i].first || (new_cost == dp[i].first
                && new_cnt > dp[i].second)) {
                dp[i] = {new_cost, new_cnt};
            }
        }
    }
    return dp[n];
}

/*
USAGE EXAMPLE:

signed main() {
    n = 1000;
    int K = 50; // We need exactly K partitions

    // Binary search for the optimal penalty `lambda`
    // The range depends on the maximum possible cost
    // difference
    long long l = -1e12, r = 1e12;
    long long ans = -1;

    while (l <= r) {
        long long mid = l + (r - l) / 2;

        pair<long long, int> res = solve_dp(mid);
        long long cost = res.first;
        int cnt = res.second;

        // If we used >= K partitions, it means the penalty is
        // small enough
        // to allow at least K partitions. We record the answer
        // and try to
        // INCREASE the penalty to force fewer partitions.
        if (cnt >= K) {
            // The real cost is the DP cost MINUS the penalty we
            // added for K partitions
            ans = cost - K * mid;
            l = mid + 1; // Increase penalty

```

```

        } else {
            r = mid - 1; // Decrease penalty
        }
    }

    cout << "Minimum cost using exactly " << K << "
    partitions: " << ans << "\n";

    return 0;
}
*/

```